: MODULE - 1 :

* Introduction to Artifical Intelligence :

[*] Artifical Intelligence :

* In 2004, John MC carthy defiened AI as Science and engineering of Making intelligent Machines, especically intelligent machines, programs.
* It was coined the birth of AI was marked by Alan Turing's Seminal work, 'computing Machinery and Intelligence" in 1950.
* Alan- Turing is known as father of AI/computer Science. continuing with his work, he proposed 'Turing test' where human interrogator would try to distinguish between computer and human text response.

▸ From a layman's Vieco, AI simply means the intelligence demonstrated by Machine that help them to mimic actions of humans, just learn from expirence, adjust to new inputs.

▸ from researcher's View, AI is a Set of Algorithim that generates results without having to be explicity instructuted to do capable of thinking and acting it

* AI can recognize objects, understand and respond to Lanuages & make decisions to solve real world problems.

[ ] How Does AI work?

* AI System work effectively with large amount of data, discover to patterns.
* AI programming focuses on three cognitive skills : learning, Reasoning and self correction.

i) Learning process :→ AI programs focus on aciquiring data & rule provide step - by step instructutions to complete Specific task. (Algorithim).

ii) Reasoning process :→ The Sucess of an AI program depends on Choosing the right algorithim to reach the desired outcome.

iii) Self - correction process : → AI programs are designed to continually enhance their alogorithim to provide the most accurate results.

: MODULE - 1 :

* Advantages and Disadvantages of Artifical Intelligence :

* Advantages :

1) perform well on tasks. that uses detailed data.
2) Takes less time to perform tasks that needs to process huge volume of data.
3) generate consistent and accurate results.
4) Can be used 24x7.
5) optimizes tasks by better utilizing resources.
6) Automates complex process
7) Minimizes downtime by predicting maintaience needs.

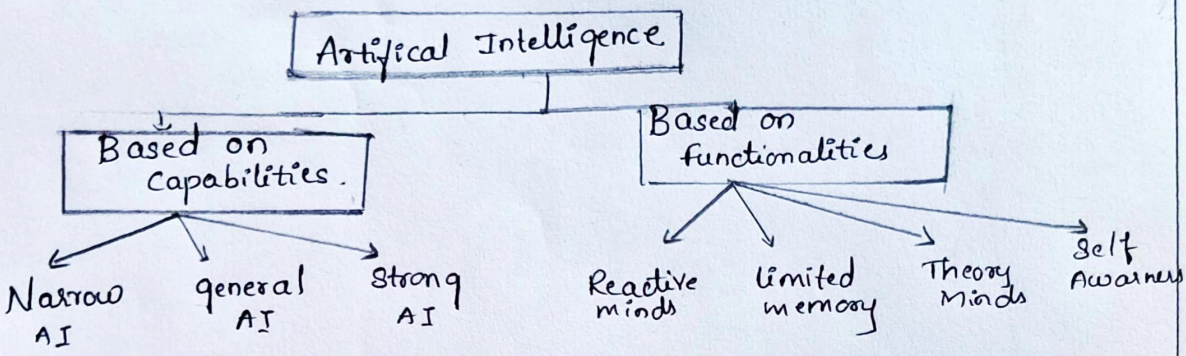
* Disadvantages :

1) Involves more cost.
2) Technical required to develop and use AI applications
3) lack of trained professionals.
4) Incomplete or inaccurate data result in disastrous results.
5) lack of capability to generate tasks.

* History of Artifical Intelligence :

1950 Turning test The turning test determing the intelligence of Machine	1955 Term AI. It is used for first time.	1966 Eliza Eliza, one of the first chatbots stimulates as psychotherapist.	1970-1980s AI winter After boom, early research.
1997 Deep Blue chess computer Blue wins against chess world champion.	2011 SIRI. Siri, is a intelligent Speech assistant, is two human rivals in quiz (jeopardy)	2011 Watson Supercomputer watson wins against two human rivals in Quiz shows.	
2014. Alexa Virtual assistant Alexa from Amazon moves into countries homes.	2016 Tay chatbox tay had to be reissued after a short time became of racist and sexist remarks.	2017 Alphago google's Alpha go beats the world champion in the game go	2018 Ethic european Union establi-shes guidelines with ethics AI.

* Types of Artifical Intelligence:



[*] Narrow AI [Weak AI]:

* Weak AI is also known as Narrow AI.
* It specifically designed to perform a Specific type of task. for.eg. Siri and Alexa are weak AI System.
* When you instruct Alexa to play a Song, responds by playing that Song.
* Weak AI System operate within limited Context and are the Most Sucessful relalization of AI to it.
* Applications in Social benefits, Google Search, Image recognition, IBM's watson are examples.

[*]. Strong AI:

* It is known as Artifical general Intelligence. or Artifical Super Intelligence (ASI). makes full attempt to resembles the human brain
* Utilizes Cognitive-Skills and logic to perform tasks for trained.
* Needs capability, Speech recognition, decision Making.
* General AI only Si-fi movies. to perform complex tasks.
* ASI will Surpass intelligence and ability of Human brain.

Weak AI	Strong AI
* It Supports a narrow range of Applications with limited Scope.	* It Support wider range of Applications with wide Scope.
* Application is good at Specific tasks.	* Incredible human-level Intelligence
* It uses Supervised and unsupervised data,	* It associates to process data
Eg: Siri, Alexa	• Eg: Advanced robotics

Reactive - Machines:→

* It as very basic machines with no memory to store and thus use past experience to determine future actions.
* IBM's and deep Blue, defeated chess grandmaster Kasporov, as fgl.
* AI System sees the pieces of chess board and responds without referring to experiences.
* Since, it lacks capabilities to store past experience reactive Machines.
* It perform only a limited number of specialized tasks. Such Machines are more trustworthy every time.
* AI application could not pursue future potential moves by opponent in better position.
* Google's AlphaGo is Another examples of game playing reactive Machine Application evaluates using Network.

Limited Memory:→

* It machines retains data only for short period of time, limited time.

1> Reinforcement learning: It enables continous learning through repeated trial and error to make better predictions.

2> Long Short term Memory (LSTM): Use data to help predict the next item in sequence.

3> Evolutionary generative Adversarial Network (E-GAN): The Model evolves over time to explore new ways in it.

Theory of Mind:

* It focuses on imitating the human brain.
* For this, forms representation about world, starting form thoughts about emotions and Memories.
* This Machine understand thoughts, emotions affect the behaviour of one's self. (making decision).

Self Awareness:

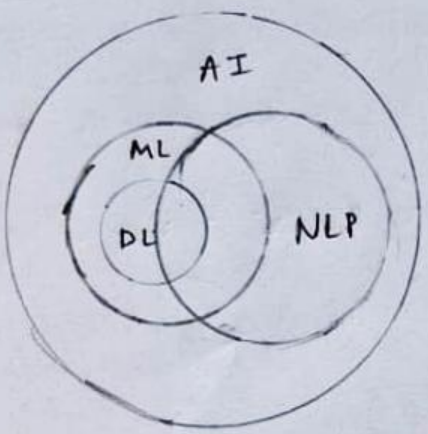
* It is estabilished, self awareness in AI System.
* It enable them to posses human level consciousness, understand current state, Own in the world.
* Knowledge imbeded by researches and its learning of the conscious context to it.

★ Is Artifical Intelligence and Cognitive computing are Same ?

- Argumented Intelligence is weak AI. that Simply improves products and Services. for eg. Automatically Highlighted.
- Artifical intelligence is Strong AI is future that would far Surpasses the world, to understand and implement complex taske.
- Cognitive computing is term especially coined for products and Services that mimic arguments human through posses.

★ Machine learning:

* It is branch of computer Science that anlysis data, patterns to teach Machine, Make decisions without any human intervention
* ML algorithms learn from experience rather than instructions.



- AI – Artifical Intelligence
- ML → Machine learning
- NLP → Natural Language Processing
- DL → Deep learning.

*

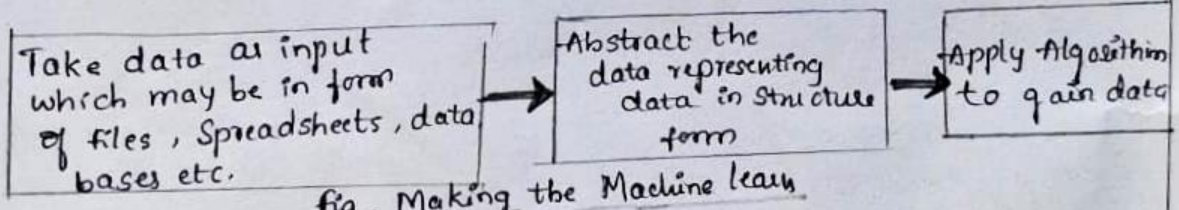


fig. Making the Machine learn

* How AI is related to Machine learning ?

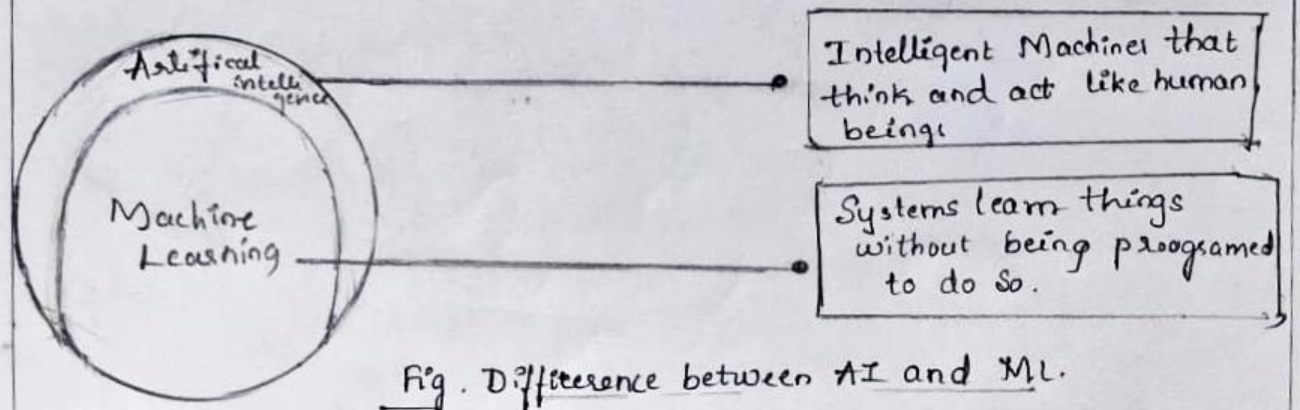


fig. Difference between AI and ML.